JR Validated Environment — MSA Module

User Guide for Measurement System Analysis

Version 1.0 | Date: 2026-03-18 | Audience: Design Engineers

1. How to Use This Guide

This guide describes the five MSA scripts available in the JR Validated Environment. Each script covers a distinct phase or approach to qualifying a measurement system. Choose the workflow that matches your test type and stage in development.

The five MSA scripts cover three workflows:

- Before using a gauge for continuous measurements: Type 1 study → Gauge R&R → Linearity & Bias
- For destructive/semi-destructive testing: Nested Gauge R&R
- For attribute (pass/fail) inspection systems: Attribute Agreement Analysis

INFO: Run Type 1 first (Section 2.3) before a full Gauge R&R (Section 2.1). Type 1 uses a single reference part and is quick to run. Gauge R&R requires multiple operators and is a larger study.

2. Quick Reference Table

Find the row that matches what you are trying to do, then go to the section listed.

I want to…	Script	Section
Qualify a continuous gauge — full study (operators × parts × replicates)	jrc_msa_gauge_rr	2.1
Qualify a gauge for destructive testing	jrc_msa_nested_grr	2.2
Quick gauge capability check (single reference part)	jrc_msa_type1	2.3
Assess gauge accuracy across its measurement range	jrc_msa_linearity_bias	2.4
Qualify an attribute (pass/fail) inspection system	jrc_msa_attribute	2.5

3. Understanding the Results

3.1 %GRR (Gauge R&R and Nested GRR)

%GRR tells you what fraction of the total observed variation comes from the measurement system itself. Industry rule: < 10% Acceptable, 10–30% Marginal, > 30% Unacceptable. The ndc (number of distinct categories) must be ≥ 5.

3.2 Cg and Cgk (Type 1 study)

Cg measures the gauge's spread relative to the tolerance. Cgk adjusts for bias. Both must be ≥ 1.33. If Cgk < Cg, the gauge has a systematic offset — investigate and re-centre before proceeding.

3.3 Linearity and Bias

A flat regression line (slope ≈ 0) means the gauge bias is consistent across its range. A significant slope means the gauge reads high at one end and low at the other — it must be re-calibrated or its use restricted to a narrower range.

3.4 Kappa (Attribute Agreement)

Kappa measures inter-rater agreement beyond chance. ≥ 0.9 Acceptable, 0.7–0.9 Marginal, < 0.7 Unacceptable. Fleiss' Kappa covers all appraisers simultaneously; Cohen's Kappa covers each appraiser internally (trial 1 vs trial 2) or vs reference.

4. Per-Script Reference

4.1 jrc_msa_gauge_rr — Standard Gauge R&R

When do I use this?

Use when the same physical parts can be measured by all operators. This is the standard Gauge R&R study: multiple operators each measure the same set of parts multiple times. Use this before relying on a measurement system for design verification.

What do I need?

Argument	What to provide
data.csv	CSV with columns: part, operator, value. Design must be balanced: every part-operator combination must have the same number of replicates (typically 2–3). Minimum: 2 parts, 2 operators, 2 replicates per cell.
--tolerance <value>	Optional. Process tolerance (USL − LSL). When supplied, %GRR is also expressed as a fraction of the tolerance width.

Example

```
jrc_msa_gauge_rr ~/msa/gauge_study.csv --tolerance 0.50
```

What to look for in the output

Check %GRR in the Verdict section. Look for %GRR < 10% (Acceptable) or < 30% (Marginal). The ndc (number of distinct categories) must be ≥ 5. The four-panel PNG shows components of variation, by-part, by-operator, and an operator×part interaction plot — a flat interaction plot means operator differences are consistent across all parts.

4.2 jrc_msa_nested_grr — Nested Gauge R&R (Destructive)

When do I use this?

Use when each operator receives their own set of specimens that cannot be shared — for example, tensile testing, chemical analysis, or any test that destroys or permanently alters the specimen.

What do I need?

Argument	What to provide
data.csv	CSV with columns: operator, part, replicate, value. Parts are nested within operators — each part belongs to exactly one operator. Part IDs may be reused across operators (operator A's part 1 and operator B's part 1 are different physical specimens). Each operator must have the same number of parts and each operator-part cell must have the same number of replicates (≥ 2).
--tolerance <value>	Optional. Process tolerance (USL − LSL).

Example

```
jrc_msa_nested_grr ~/msa/destructive_study.csv --tolerance 10.0
```

What to look for in the output

Interpret %GRR the same way as a standard Gauge R&R (< 10% Acceptable, < 30% Marginal). Note: because parts are not shared, the part-to-part estimate is confounded with the operator batch — the analysis assumes all operators received specimens from the same population.

WARNING: Nested GRR limitation: Because parts are not crossed across operators, part-to-part variation and reproducibility are partially aliased. The %GRR estimate is conservative. If parts can physically be shared without being destroyed, run jrc_msa_gauge_rr instead.

4.3 jrc_msa_type1 — Type 1 Gauge Study

When do I use this?

Use for a quick initial gauge capability check before committing to a full Gauge R&R study. A single reference part of known value is measured repeatedly under the same conditions. The study quantifies the gauge's spread (Cg) and any systematic bias (Cgk).

What do I need?

Argument	What to provide
data.csv	CSV with a 'value' column. Optionally an 'id' column for the run chart x-axis. Minimum 10 measurements; 25–50 are typical.
--reference <value>	Required. The known true value of the reference part (e.g. from a certified reference artefact).
--tolerance <value>	Required. The process tolerance (USL − LSL) the gauge is intended to support.

Example

```
jrc_msa_type1 ~/msa/type1_study.csv --reference 50.0 --tolerance 1.0
```

What to look for in the output

Cg ≥ 1.33 and Cgk ≥ 1.33 are required. If Cg ≥ 1.33 but Cgk < 1.33, the gauge is precise but offset — check calibration. If both are below 1.33, the gauge variability is too large for this tolerance. The run chart shows whether variability is stable over time; the histogram shows whether it follows a normal distribution.

4.4 jrc_msa_linearity_bias — Linearity and Bias

When do I use this?

Use to check whether a gauge has a consistent bias across its measurement range, or whether its accuracy changes at low vs high values. Run after Type 1 and Gauge R&R for a complete MSA picture.

What do I need?

Argument	What to provide
data.csv	CSV with columns: part, reference, value. Each part must have a fixed reference value. At least 2 (ideally 5+) distinct reference levels covering the gauge's operating range. Multiple replicates per reference level are recommended.
--tolerance <value>	Optional. Process tolerance.

Example

```
jrc_msa_linearity_bias ~/msa/linearity_study.csv --tolerance 5.0
```

What to look for in the output

A non-significant slope (p > 0.05) means bias is consistent across the range — acceptable. A significant slope means the gauge reads differently at different reference levels — investigate calibration or restrict the gauge's working range. Per-part bias bars show which reference levels have the largest offset.

4.5 jrc_msa_attribute — Attribute Agreement Analysis

When do I use this?

Use to qualify an inspection system that produces categorical results (Pass/Fail, Grade A/B/C, Accept/Reject). Multiple appraisers each inspect the same set of parts multiple times.

What do I need?

Argument	What to provide
data.csv	CSV with columns: part, appraiser, trial, rating. The design must be balanced: every appraiser-part combination must have the same number of trials (≥ 2). The 'rating' column contains the categorical result (any text or number). Optionally include a 'reference' column with the known correct rating for each part — this enables the vs-reference Kappa calculation.

Examples

```
jrc_msa_attribute ~/msa/attribute_study.csv
jrc_msa_attribute ~/msa/attribute_study_with_ref.csv
```

What to look for in the output

Within-appraiser Kappa ≥ 0.9 means each appraiser is consistent with themselves. Between-appraiser Fleiss' Kappa ≥ 0.9 means the inspection system is consistent across operators. If a reference column is present, each appraiser's Kappa vs reference measures accuracy (not just consistency). Verdict: ≥ 0.9 Acceptable, 0.7–0.9 Marginal, < 0.7 Unacceptable.

TIP: The reference column should contain the 'ground truth' rating — for example, ratings from a subject matter expert or certified inspector. Without it, the analysis can only tell you whether appraisers agree with each other, not whether they are correct.

5. Preparing Your Data

Each script expects a CSV file. The table below lists the required and optional columns for each script.

Script	Required columns	Optional columns
jrc_msa_gauge_rr	part, operator, value	—
jrc_msa_nested_grr	operator, part, replicate, value	—
jrc_msa_type1	value	id
jrc_msa_linearity_bias	part, reference, value	—
jrc_msa_attribute	part, appraiser, trial, rating	reference

Column names are case-insensitive. Extra columns are ignored. Numeric values may use European decimal commas (they are automatically converted). The 'rating' column in jrc_msa_attribute may contain any text or number.

6. Recommended Study Workflow

Follow this sequence when qualifying a continuous measurement system for design verification:

1. Run jrc_msa_type1 — quick gauge capability screen (Cg, Cgk). If Cg < 1.33, stop: the gauge is not capable for this tolerance.
2. Run jrc_msa_gauge_rr (or jrc_msa_nested_grr for destructive testing) — full %GRR study. If %GRR > 30%, stop: the gauge must be improved.
3. Optionally run jrc_msa_linearity_bias to verify accuracy across the operating range.
4. Document all three PNG outputs in the Measurement System Qualification record in your DHF.

WARNING: Do not proceed to design verification if %GRR > 30% or Cg/Cgk < 1.33. Measurements from an unqualified gauge cannot support FDA submission claims.